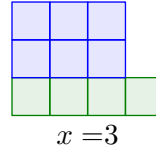
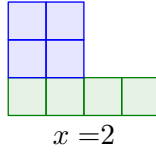
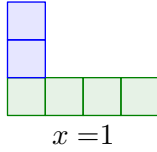


Exercises for Chapter 0

0.1.1

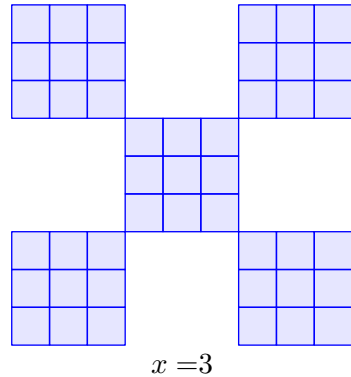
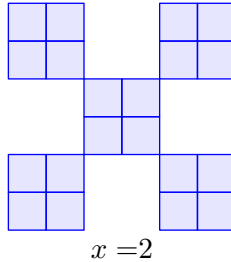
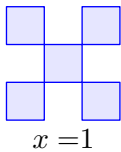
Let the number of boxes in the below figures be given by $f(x)$.



- Find an expression for $f(x)$.
- How many boxes are there when $x = 100$?
- What is x when $f(x) = 24$?

0.1.2

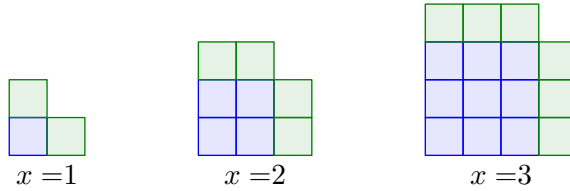
Let the number of boxes in the below figures be given by $a(x)$.



- Find an expression for $a(x)$.
- How many boxes are there when $x = 20$?
- What is x when $a(x) = 405$?

0.1.3

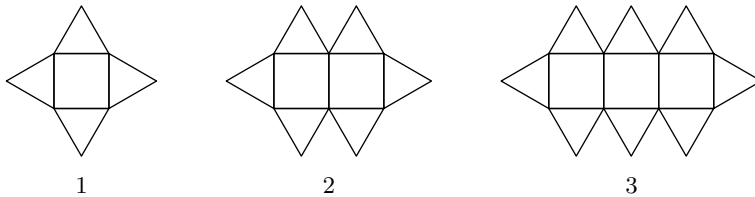
Let the number of boxes in the below figures be given by $b(x)$.



- Find an expression for $b(x)$.
- How many boxes are there when $x = 20$?
- What is x when $b(x) = 80$?

0.1.4 (EGV22D1)

The below figure shows the three first figures of a pattern. The figures are made up by triangles and squares.



How many triangles and how many squares will be present in figure number 10?

0.1.5

Let x be a positive integer.

- Make a function $p(x)$ yielding the value of the x th positive, even number.
- Make a function $o(x)$ yielding the value of the x th positive odd number.

0.2.1

Find the slope and the intercept of the function.

a) $f(x) = 5x + 10$ b) $g(x) = 3x - 12$

c) $h(x) = -\frac{1}{7}x - 9$ d) $i(x) = \frac{3}{2}x - \frac{1}{4}$

0.2.2

Draw the function on the interval $x \in [-5, 5]$:

a) $f(x) = 2x - 1$ b) $g(x) = -3x + 5$

0.3.1

Given the equation set

$$x - y = 5 \quad (\text{I})$$

$$x + y = 9 \quad (\text{II})$$

- a) Explain why the solutions of this set yield the interception point of the functions

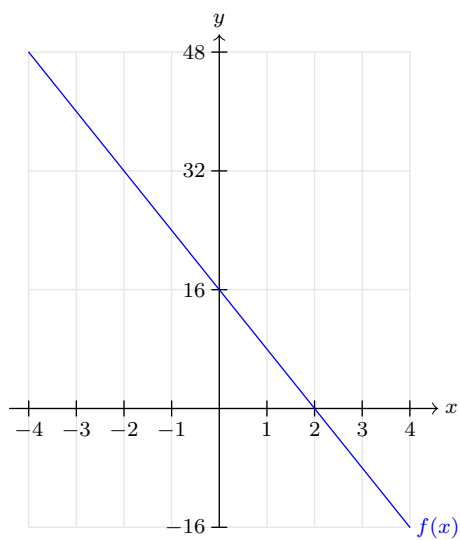
$$f(x) = x - 5$$

$$g(x) = 9 - x$$

- b) Solve the equation set.

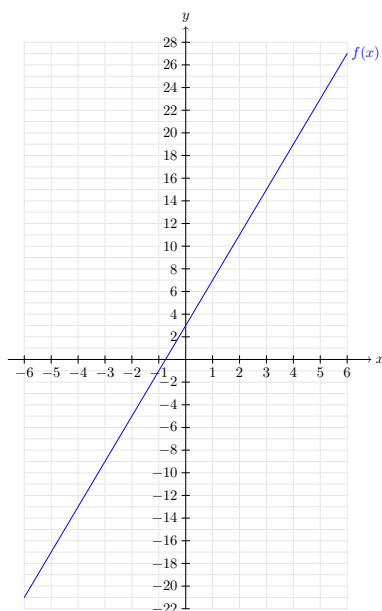
0.3.2

Find the expression of $f(x)$



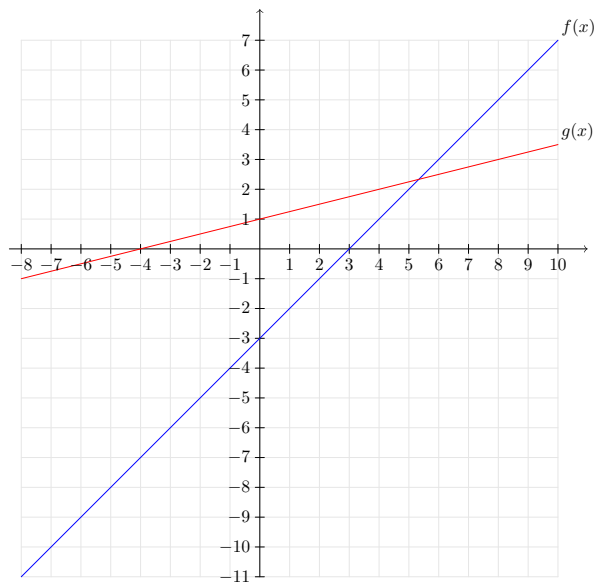
0.3.3

Find the expression of $f(x)$



0.3.4

Find the expression of $f(x)$ and $g(x)$.



Ponder 1

Ponder 2

Use the formulas from [Exercise 0.1.5](#) to prove that

- a) the sum/difference of two even numbers is an even number.
- b) sum/difference of two odd numbers is an even number.
- c) the sum/difference of an even number and an odd number is an odd number.

Ponder 3

- a) A linear function $f(x)$ has slope 3, and the point $(2, 1)$ lies on the graph of f . Find the expression of f .
- b) A linear function $f(x)$ has slope a , and the point (x_1, y_1) lies on the graph of f . Prove that

$$f(x) = a(x - x_1) + y_1$$

(This equation is called the **point-slope equation** .)

Ponder 4

Given the functions $f(x)$ and $g(x)$, where the graph of g is the line passing through $A = (a, f(a))$ and $B = (b, f(b))$. Prove that

$$f - g = f(x) - \frac{f(b) - f(a)}{b - a}(x - a) + f(a)$$